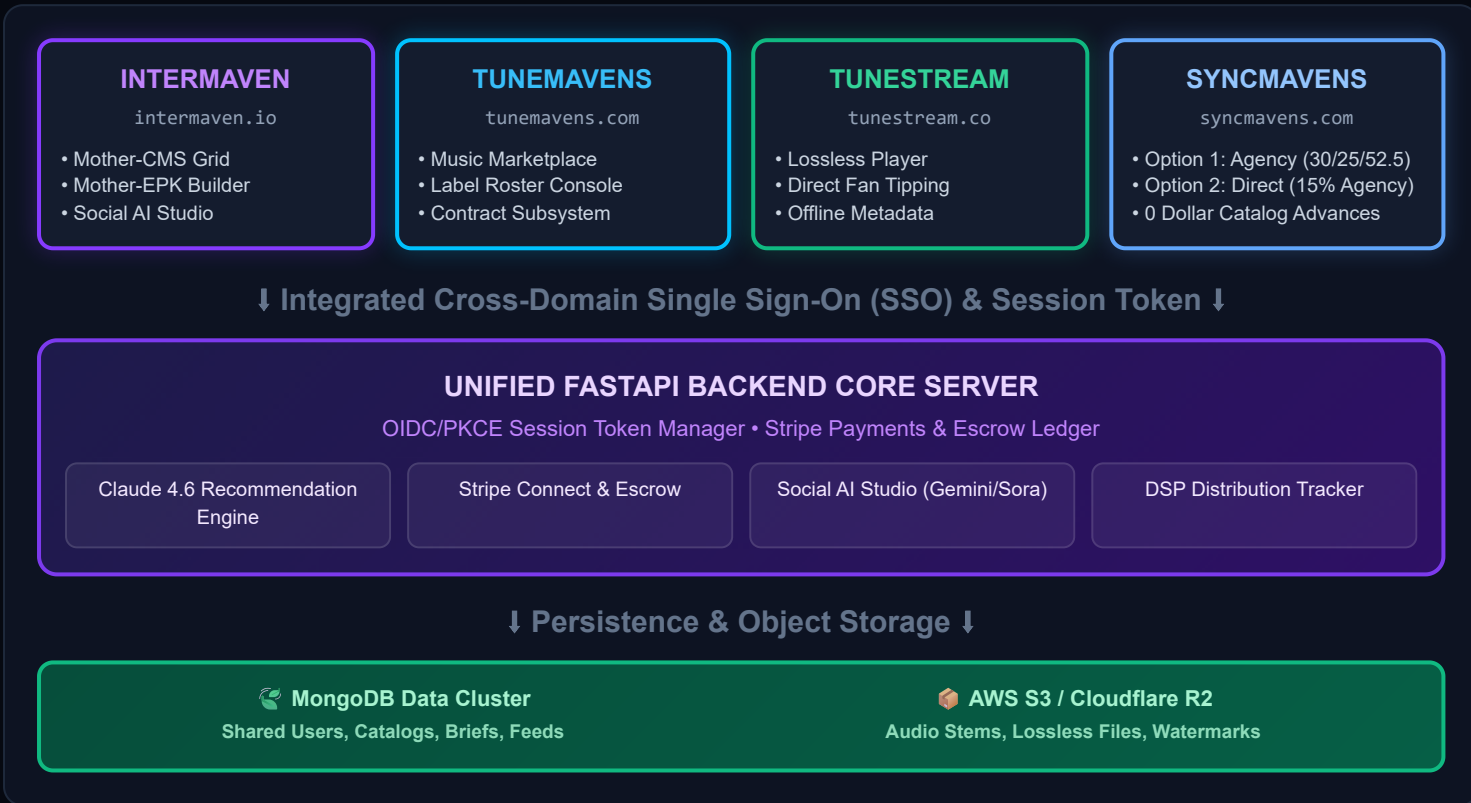


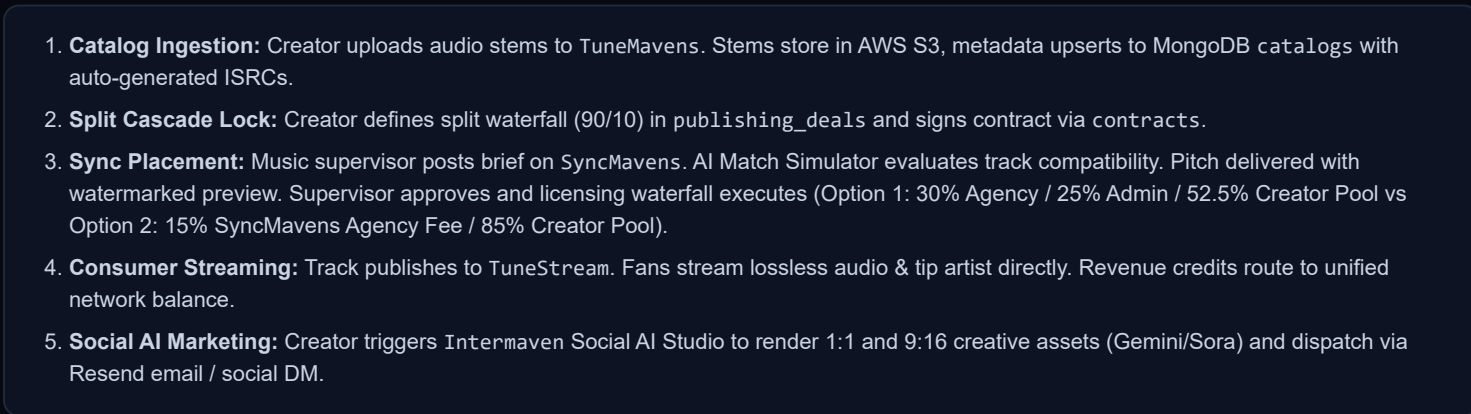
ECOSYSTEM SYSTEM ARCHITECTURE & FLOWCHART

Monorepo Technical Blueprint, Connectivity Flowchart & Data Schema Index

1. High-Impact Ecosystem Connectivity Flowchart



2. End-to-End Data & Asset Lifecycle Sequence



3. MongoDB Shared Collection Index

Collection Name	Key Fields & Attributes	Primary Consuming Apps	Business Purpose
users	_id, email, roles[], primary_role, plan, credits	All 4 Platforms	Unified SSO identity, role permissions, and credit balance
catalogs	_id, isrc, title, stems_s3_url, splits[]	TuneMavens, SyncMavens, TuneStream	Centralized metadata index for masters, stems, and audio assets
briefs	_id, supervisor_id, budget, deadline, genre	SyncMavens	Licensing opportunities submitted by music supervisors
pitches	_id, brief_id, catalog_id, match_score, waterfall_splits	SyncMavens, TuneMavens	Pitch submissions and licensing deal ledgers
publishing_deals	_id, user_id, tier, splits_cascade, status	TuneMavens	Publishing agreement records and split calculation rules
contracts	_id, contract_type, locked_clauses[], signatures[]	TuneMavens, SyncMavens	E-signature and legal contract negotiation state machine
campaigns	_id, name, segment, channels[], stats	Intermaven, TuneMavens	Multi-channel CRM growth marketing campaigns

4. Detailed Technical Stack Specifications & Monorepo Architecture

4.1 Monorepo Directory Layout & Application Boundaries

tunemaven/	
├─ apps/	
│ ├─ portal/	# Main Portal & Dashboard Hub (tunemavens.com)
│ ├─ tunestream/	# Consumer Audio Streaming App (tunestream.co)
│ └─ syncmavens/	# Sync Licensing Marketplace (syncmavens.com)
├─ packages/	
│ ├─ shared-ui/	# Shared CSS Tokens, UI Components & Modals
│ └─ shared-auth/	# Shared OIDC/PKCE Session Helpers
├─ backend/	
│ ├─ server.py	# Primary FastAPI Application Entry
│ ├─ auth.py	# PyJWT Cross-Domain Token Validation
│ ├─ routes/	# Domain Routers (sso, match, stream, crm, cms)
│ └─ models.py	# Pydantic v2 Data Transfer Models
└─ deploy/	# Multi-Domain Hostinger VPS Nginx Configs

4.2 Layer-by-Layer Technical Specification Matrix

Tech Layer	Platform Standard	Implementation & Package Specification
Frontend Framework	React 18 + Vite	Modular Monorepo Monolith with Isolated Build Targets (dist/portal, dist/tunestream, dist/syncmavens).
Styling System	Vanilla CSS + Tokens	Custom HSL design tokens, glassmorphism, responsive grid layouts, Outfit/Inter fonts. Zero CSS framework bloat.
Native Packaging	Capacitor	Wraps web frontend applications for iOS & Android native deployment with native audio drivers.
Backend Engine	FastAPI (Python 3.11)	ASGI asynchronous web server running Uvicorn workers with non-blocking event loops.
Database Layer	MongoDB + Motor	Multi-tenant MongoDB replica set cluster using Motor for async Python database I/O.
Object Storage	AWS S3 / Cloudflare R2	Direct browser presigned URL upload pipeline for lossless audio stems (WAV/AIFF) and preview clips.
Single Sign-On	OIDC + PKCE	Custom OAuth2/OIDC SSO server (sso_router.py) issuing PyJWT cookies across .tunemavens.com.
Payment Gateway	Stripe Connect	Direct payouts, 4-tier entitlements, QR event ticketing scanner, Option 1 (30/25/52.5) & Option 2 (15% agency fee) ledgers.

Tech Layer	Platform Standard	Implementation & Package Specification
AI Recommendation	Claude Sonnet 4.6	15s capped recommendation synthesis engine (<code>users_router.py</code>) recommending tools and apps.
Social AI Studio	Gemini Nano + Sora 2	Multi-format image (<code>/generate-art</code>) and short-form video clip (<code>/generate-video</code>) campaign generator.
CRM Engine	Resend API + Inbox	Multi-channel messaging engine dispatching emails via Resend and in-app inbox threads (<code>messages</code>).